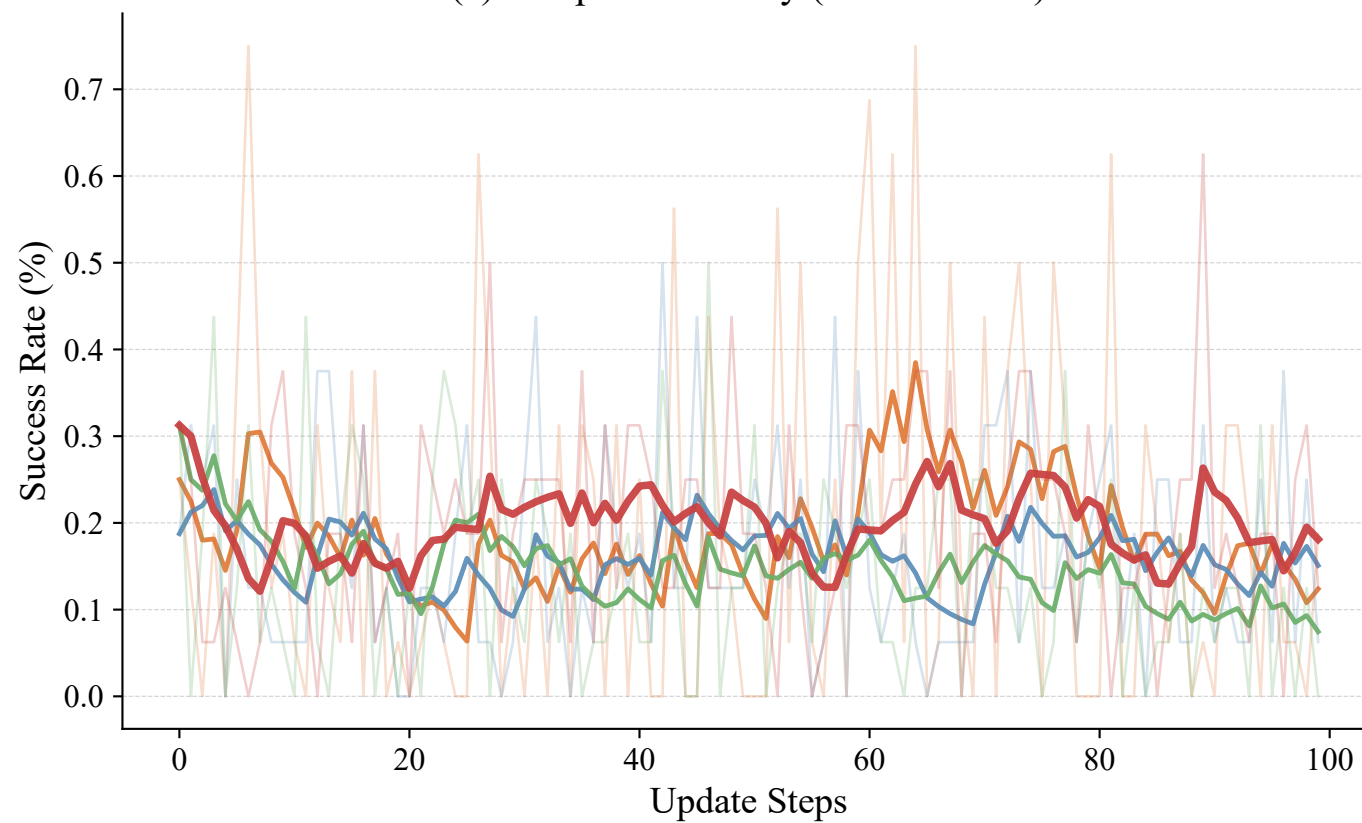


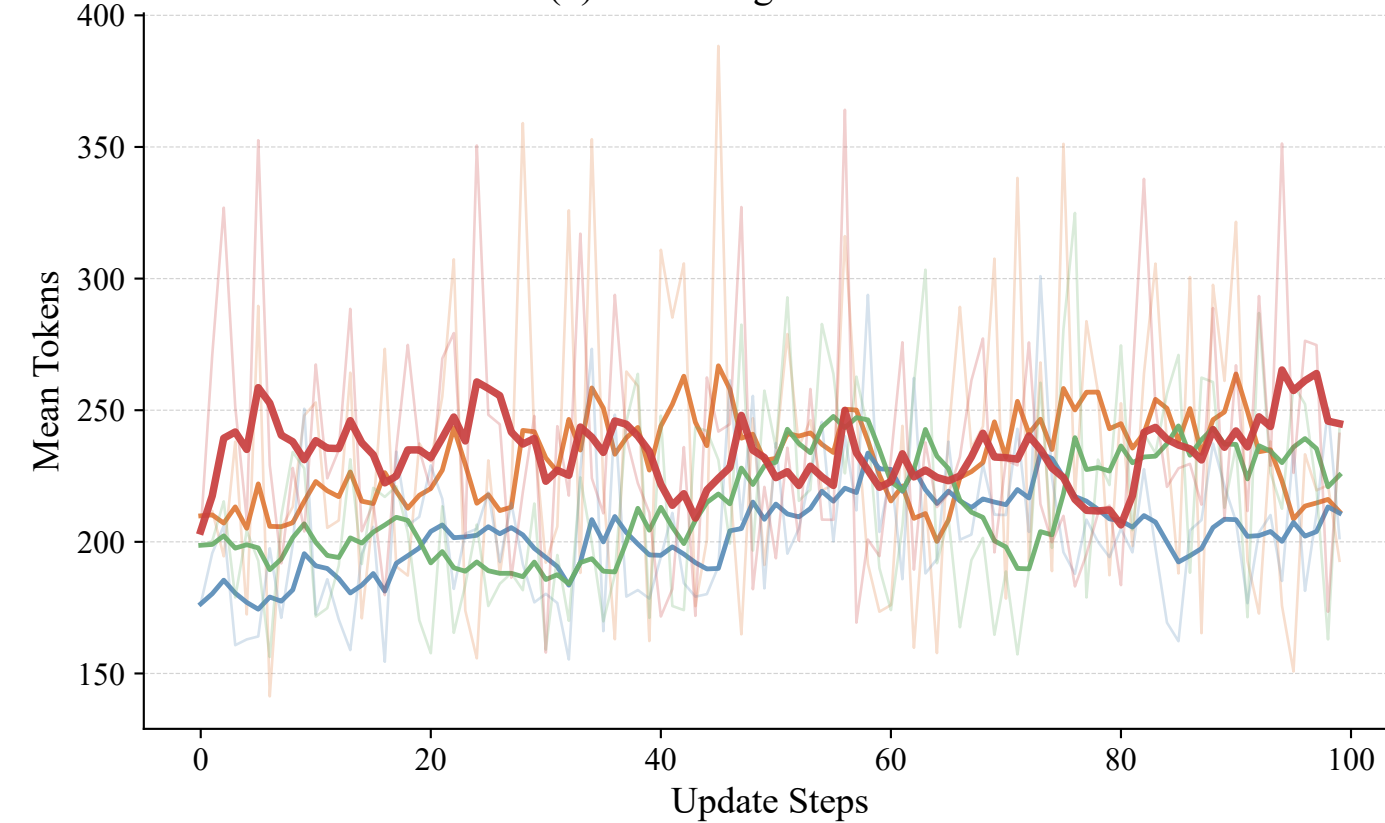
GRPO Group Size (G) Ablation: Scaling Behaviors (EBS=16/64)



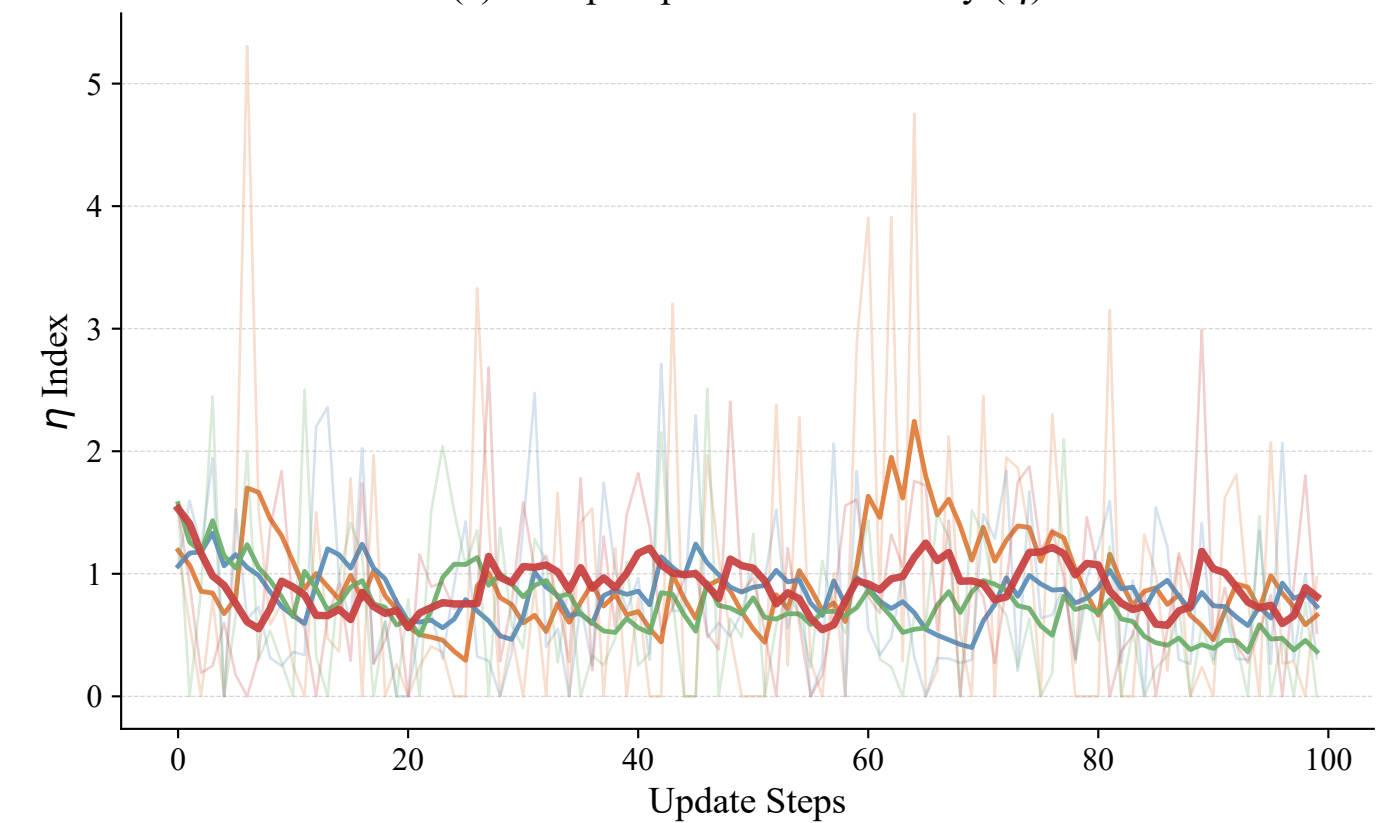
(a) Sample Efficiency (Success Rate)



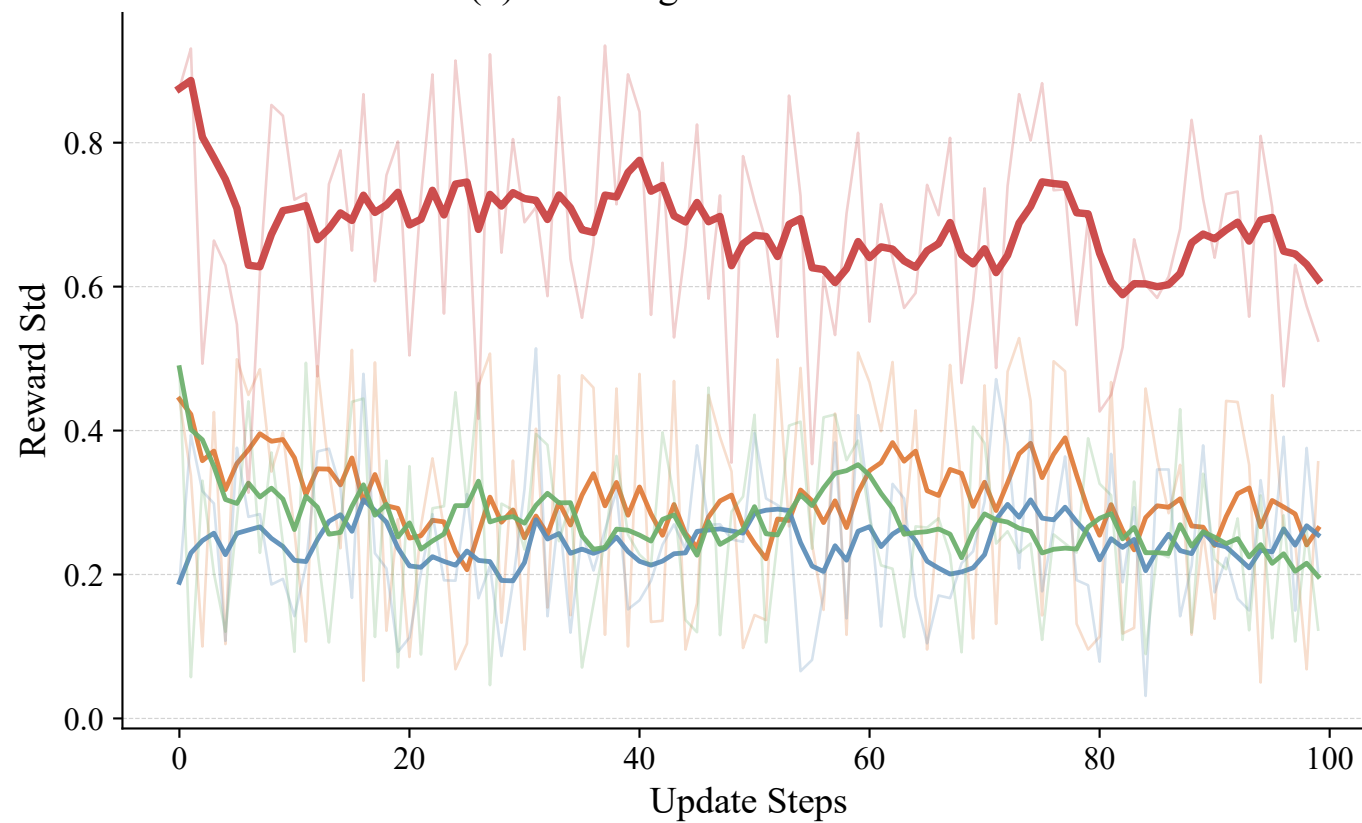
(b) Reasoning Tokens vs. G



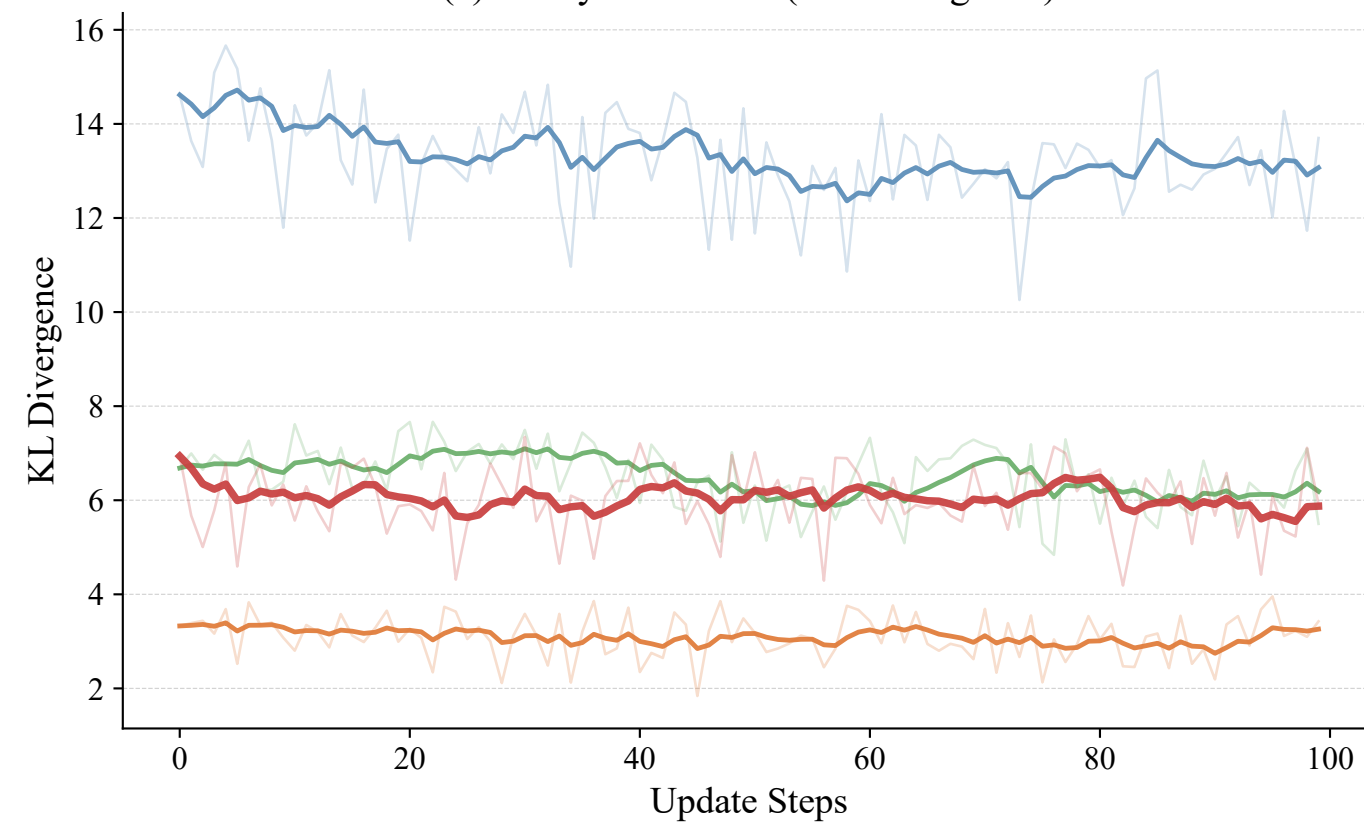
(c) Group Exploration Efficiency (η)



(d) Advantage Estimation Variance



(e) Policy Constraint (KL Divergence)



(f) Gradient Stability (L2 Norm)

